SMART SEP

AI & IoT-Powered Advanced Biodegradable and Non-Biodegradable Waste Segregation and Monitoring System

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Objective & Extended Abstract (Problem Definition)

The Global Challenge: Over 2 billion tons of municipal waste is generated annually, with a projected 70% increase by 2050.

The Gap: Many cities collect only 50–80% of waste despite spending up to 95% of budgets on collection. Manual sorting is labor-intensive, hazardous, and leads to high contamination in recycling streams.

Our Objective: To provide a comprehensive IoT-enabled system that automates outdoor monitoring, real-time classification, and robotic collection to reduce manual labor and environmental harm

Societal/Market Relevance: Prevents methane-related hazards and health risks from open dumps while optimizing municipal resources.

Uniqueness: Unlike existing stationary or indoor "smart bins," this system fuses multiple sensing modalities—vision, gas, location, and fill levels—with an autonomous robotic collection vehicle

THEORETICAL BACKGROUND

- ***Basic Principle***
- ***AI-Enhanced Edge Computing:*** Processes visual data locally on an ESP32-CAM module using OpenCV to classify waste without cloud latency.
- ***Multi-Sensor Fusion:*** Combines time-of-flight ultrasonic data, electrochemical gas sensing, and thermohydrometric data for holistic bin status.
- ***Robotic Automation:*** Employs 4-DOF (Degrees of Freedom) kinematics for adaptive waste handling and metal segregation

Mathematical & Simulation Basics

1) Ultrasonic Sensor (HC-SR04)

Distance = (Time × 343) / 2

Fill Level = Bin Height – Distance

Range: 0 cm – 25 cm

2) DC Motor (Car Movement)

Forward: M1 = 1, M2 = 0

Backward: M1 = 0, M2 = 1

Speed ∝ PWM Duty Cycle

3) Servo Motor (Robotic Arm)

$0° \le \theta \le 180°$

Angle = ((Pulse Width – 1ms) / 1ms) × 180°

4) GPS (NEO-6M)

$d = 2r \sin^{-1}(\sqrt{[\sin^2(\Delta\varphi/2) + \cos\varphi_1 \cos\varphi_2 \sin^2(\Delta\lambda/2)]})$

5) MQ-4 Gas Sensor

ppm = Sensor Output / Calibration Factor

Alert: CH_4 > 2 ppm

6) DHT11 Sensor

$20°C \le T \le 35°C$

$30\% \le H \le 70\%$

FLOWCHART

Process Flow

Flowchart Overview:

Sense: Continuous monitoring of fill levels, gas, and environmental conditions.

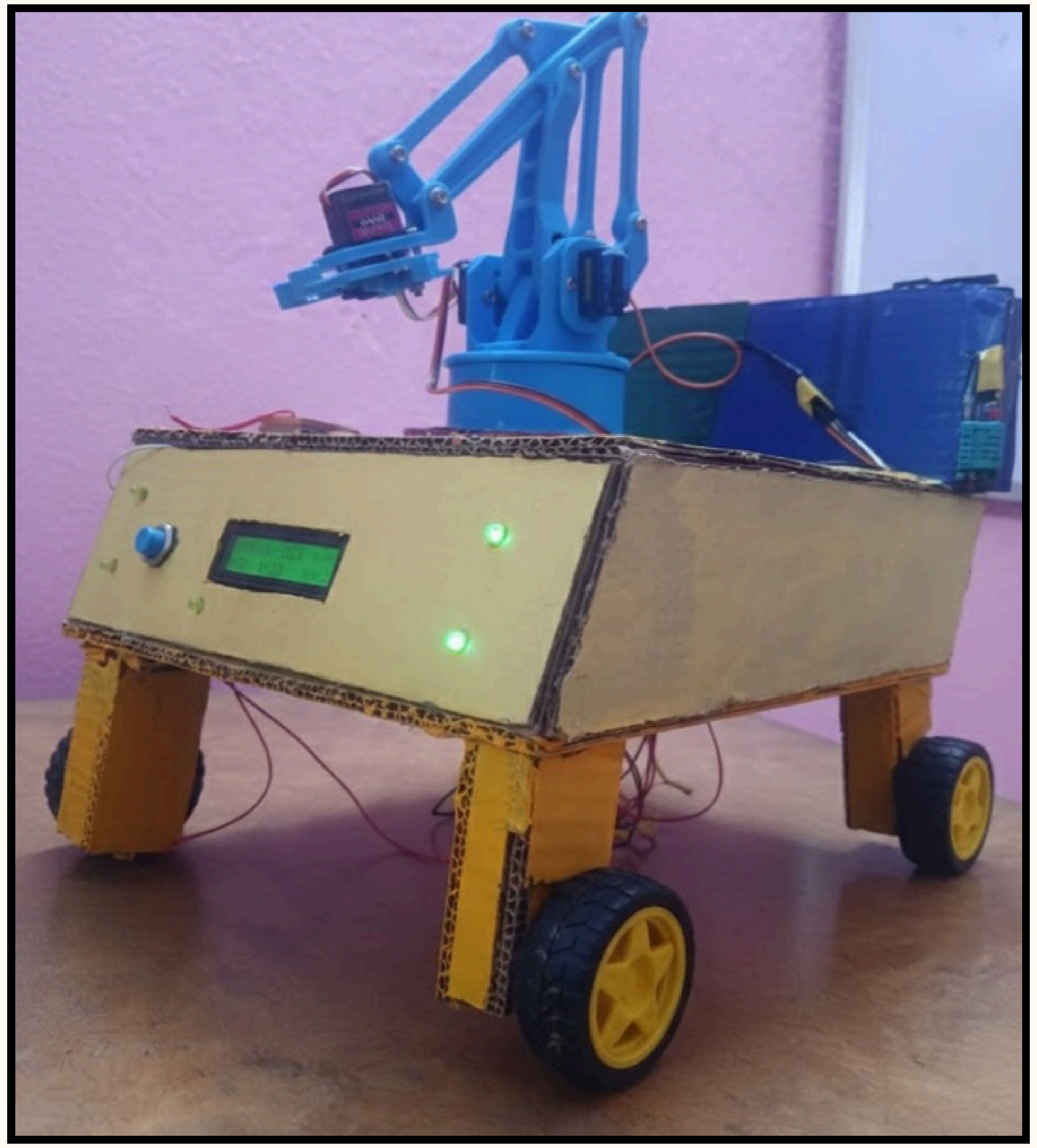
Classify: ESP32-CAM captures images; OpenCV classifies as Biodegradable/Non-Biodegradable.

Communicate: Data transmitted via IoT Gateway to Cloud; alerts sent to mobile app.

Act: Robotic arm sorts metal; Bluetooth-controlled car navigates to full bins (>95%).

SCAN ME

Our Present Productt



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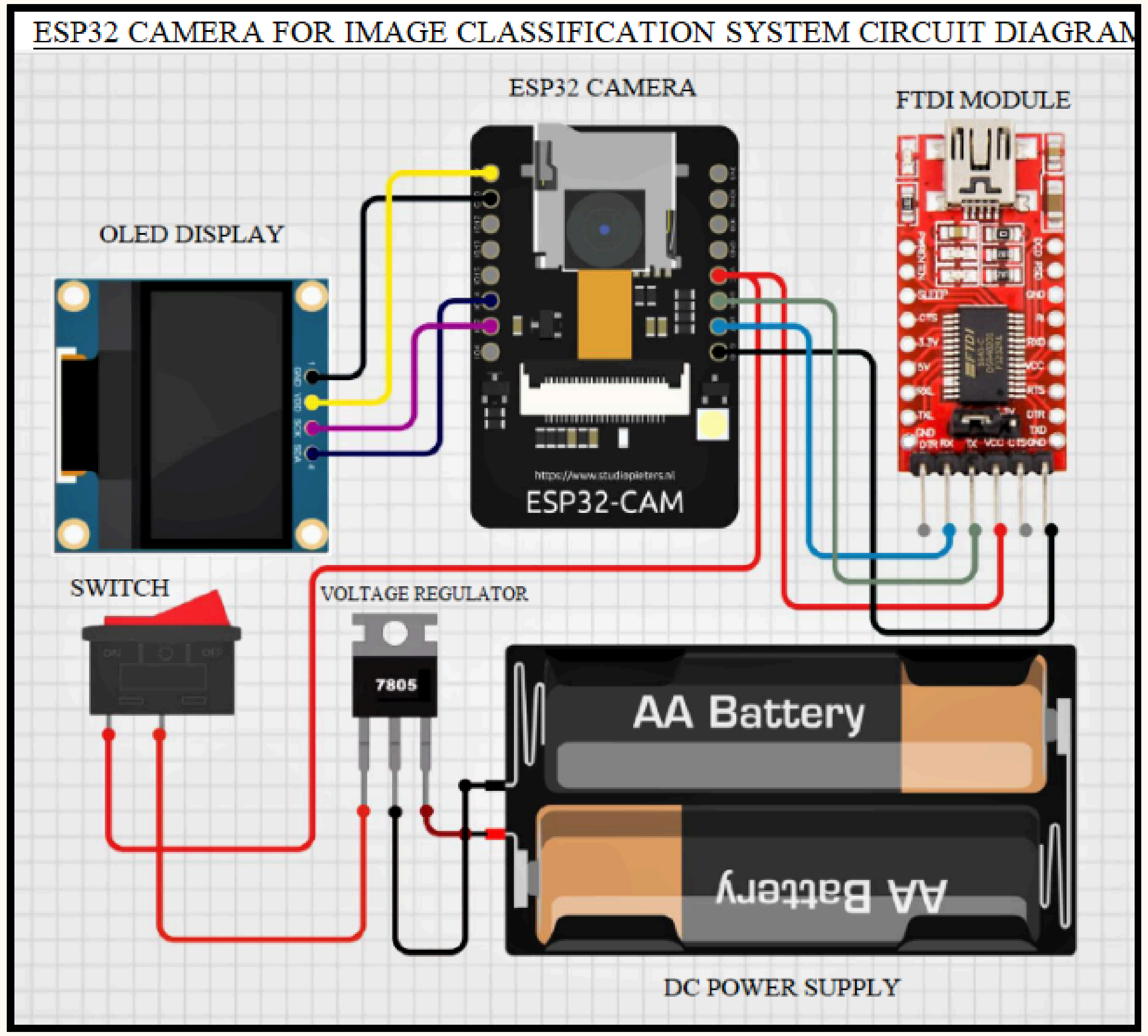
IMPLEMENTATION (HARDWARE PROTOTYPE)

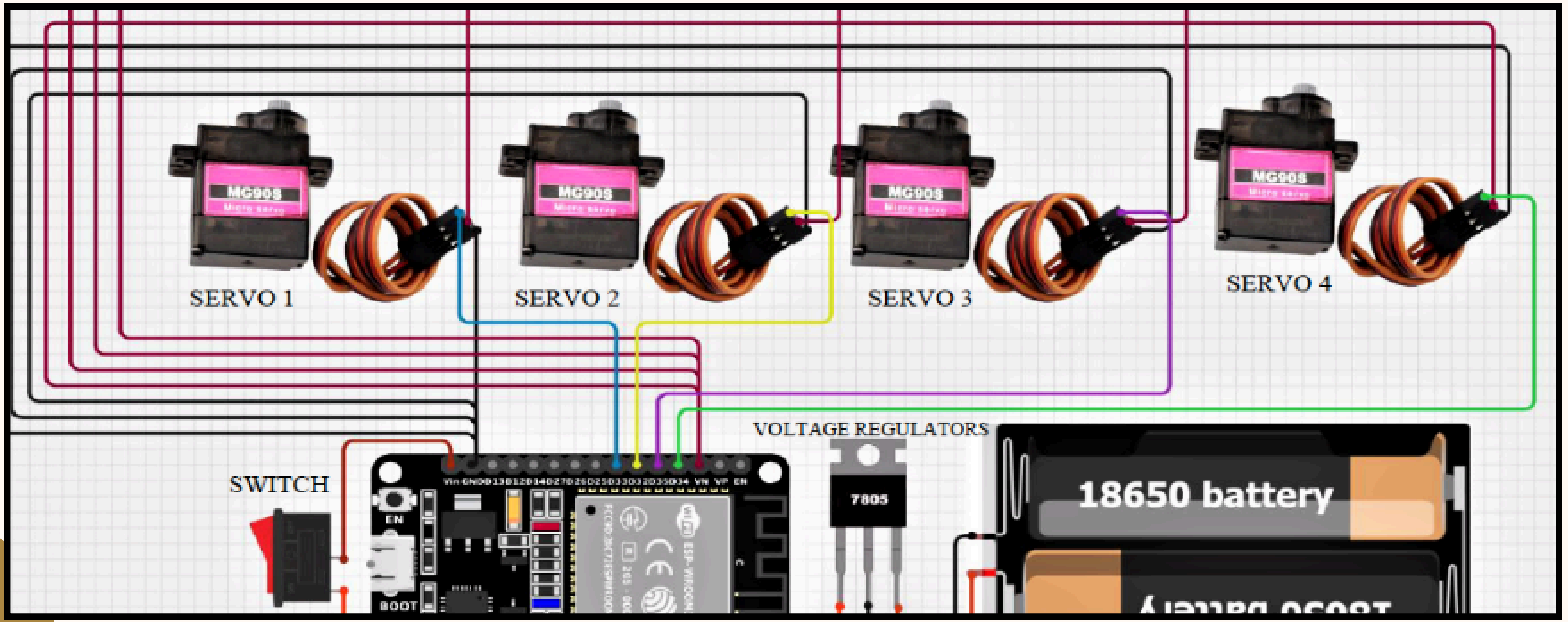
SENSOR NODES: HC-SR04 (DISTANCE), MQ-4 (METHANE), DHT11 (TEMP/HUMIDITY), METAL DETECTOR.

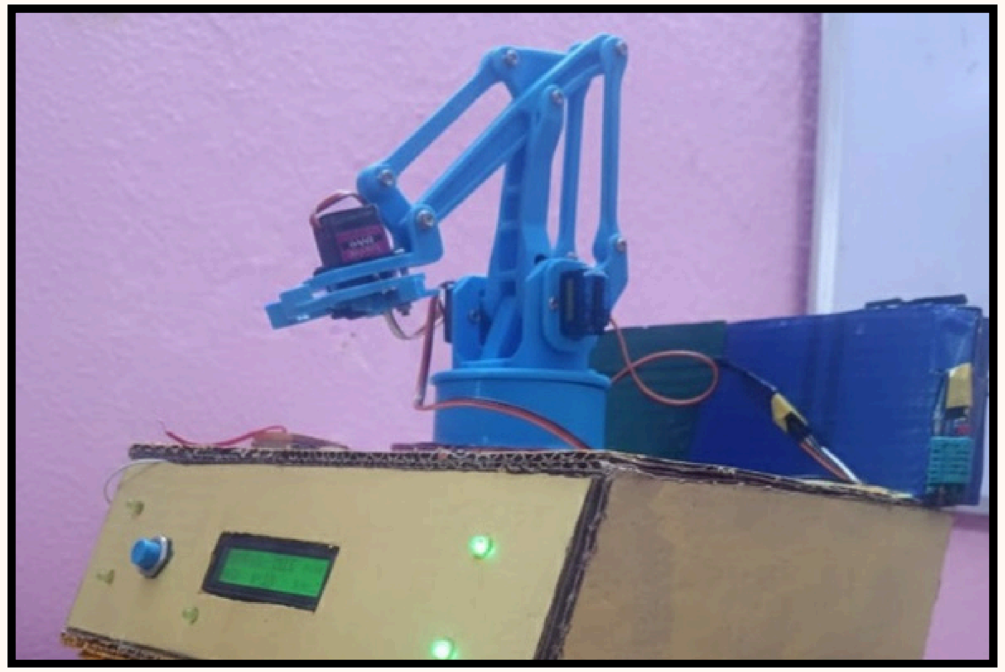
CORE CONTROLLER: ESP32-CAM FOR AI PROCESSING AND WI-FI CONNECTIVITY.

MOBILE UNIT: BLUETOOTH-CONTROLLED CAR WITH NEO-6M GPS FOR LIVE TRACKING AND A 4-DOF ROBOTIC ARM.

POWER: INTEGRATED SOLAR-POWERED BATTERY SYSTEM FOR OFF-GRID OPERATION.







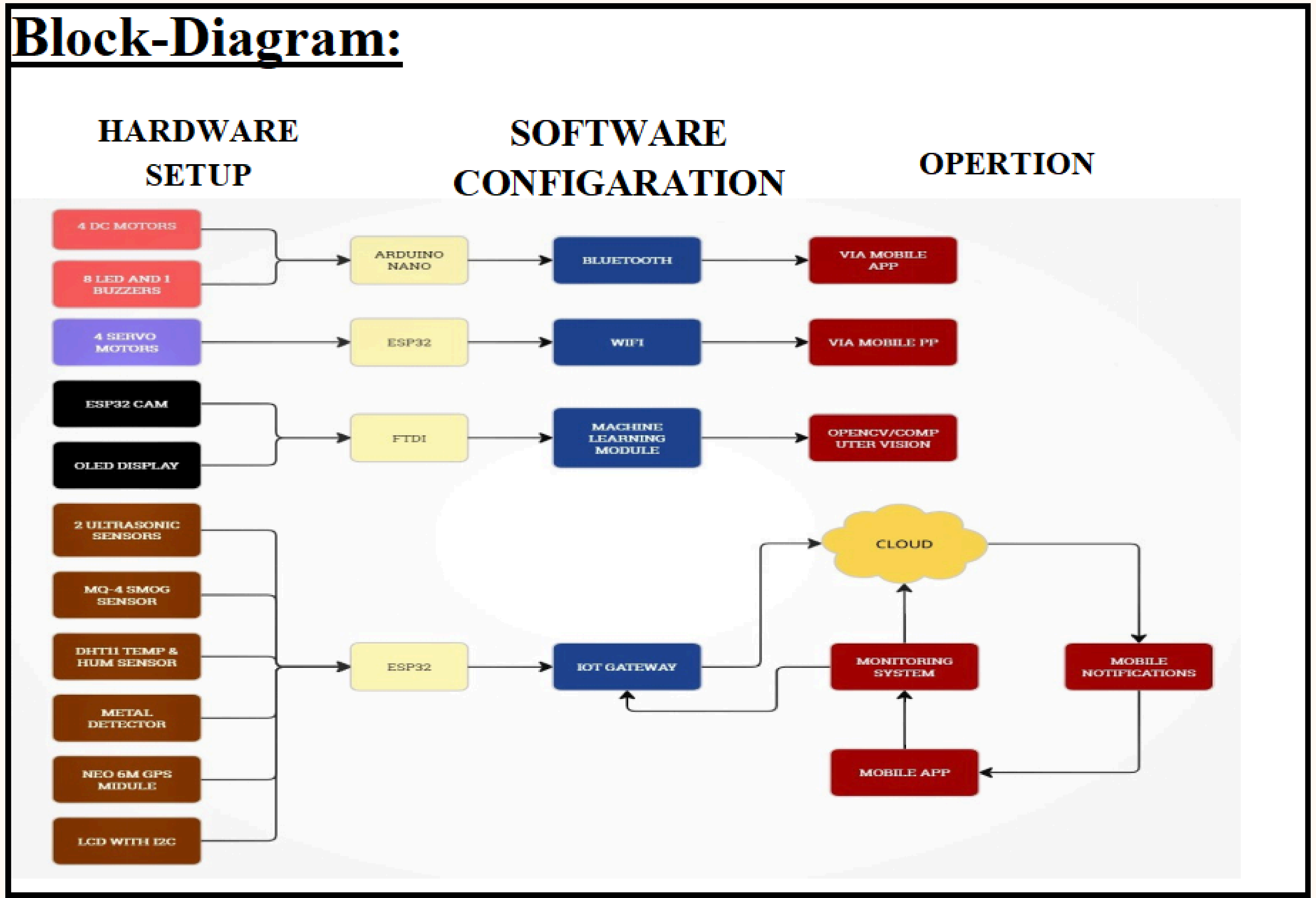
IMPLEMENTATION (SOFTWARE & CLOUD INTEGRATION)

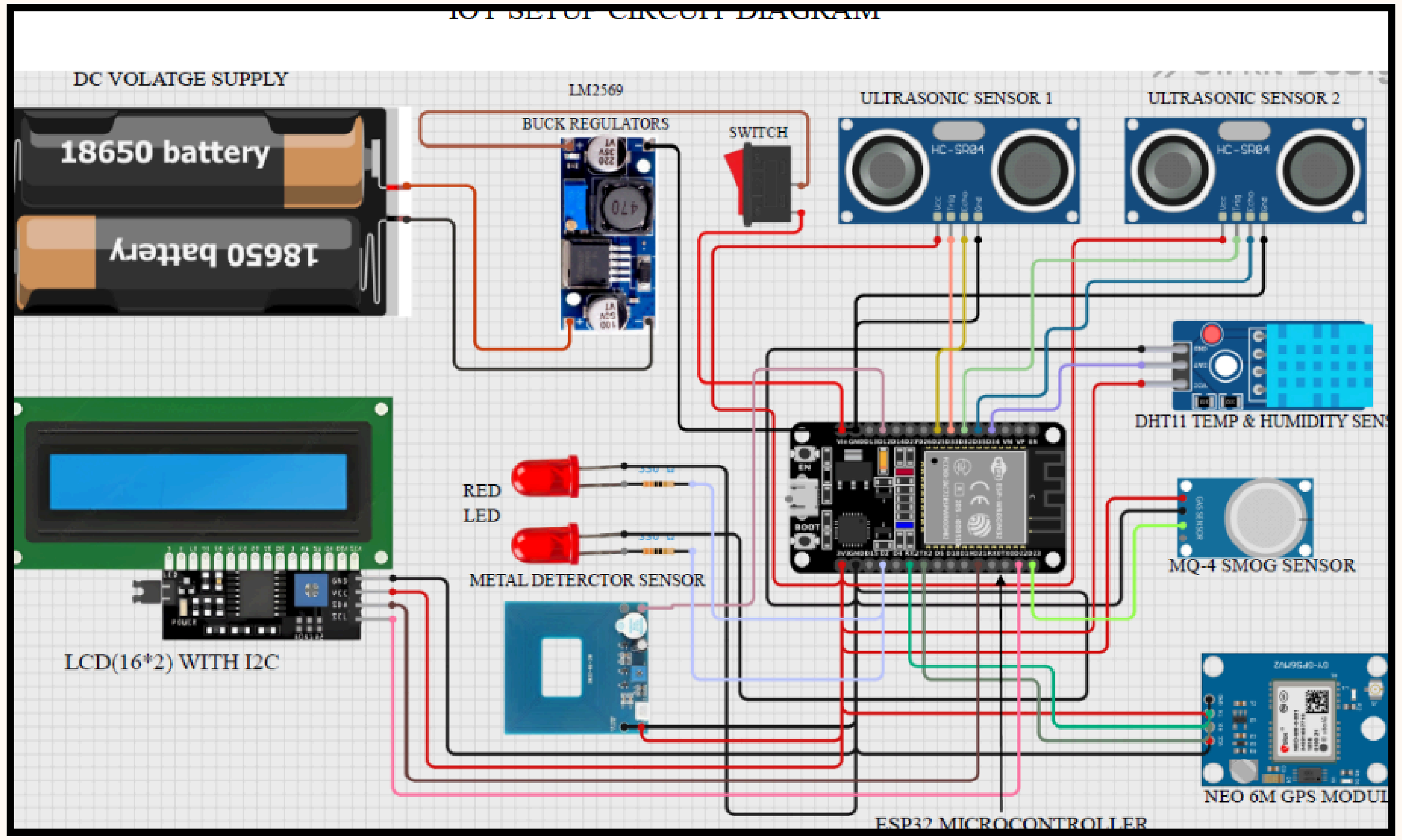
***Edge AI*:** **OpenCV Python model optimized for low-computational overhead on microcontrollers.**

***IoT Gateway*:** **Central hub for data collection, supporting multiple wireless protocols (Wi-Fi, GPS).**

***Cloud Platform*:** **Aggregates time-stamped data for audit trails and route optimization.**

***Mobile App*:** **Provides real-time visualization of GPS locations, bin status, and manual override controls.**

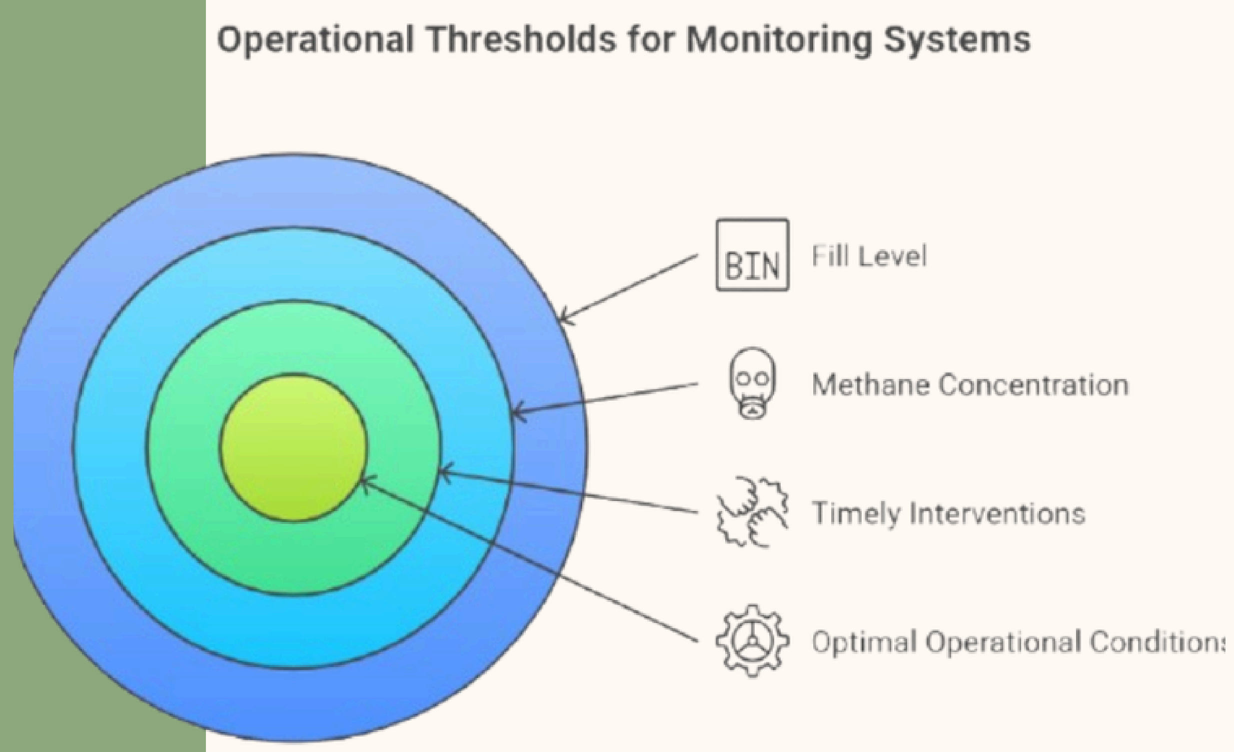


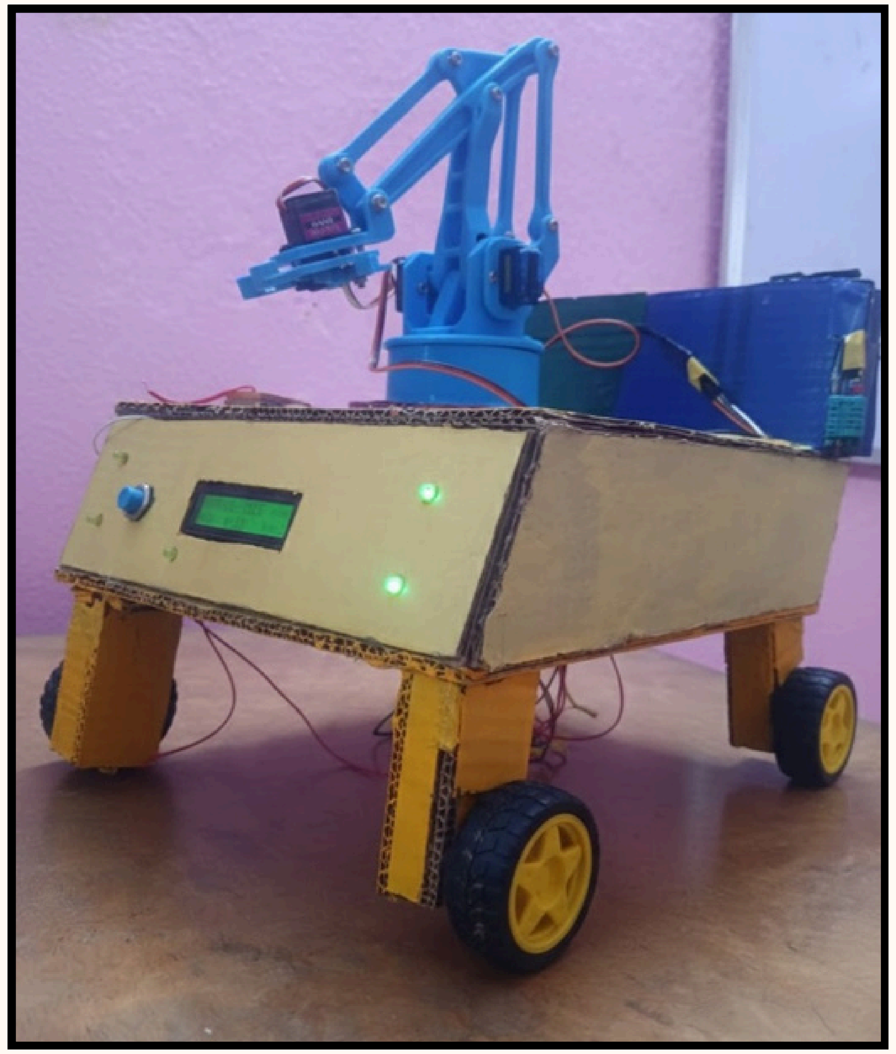


IMPLEMENTATION (EXPERIMENTAL RESULTS)

Classification Accuracy : ***Achieved 98% accuracy in distinguishing waste types under varying lighting.***
Operational Thresholds: ***Reliable alerts triggered at 95% fill level and >1500 ppm Methane.***

Responsiveness: ***Minute-by-minute updates for environmental sensing and location tracking.***





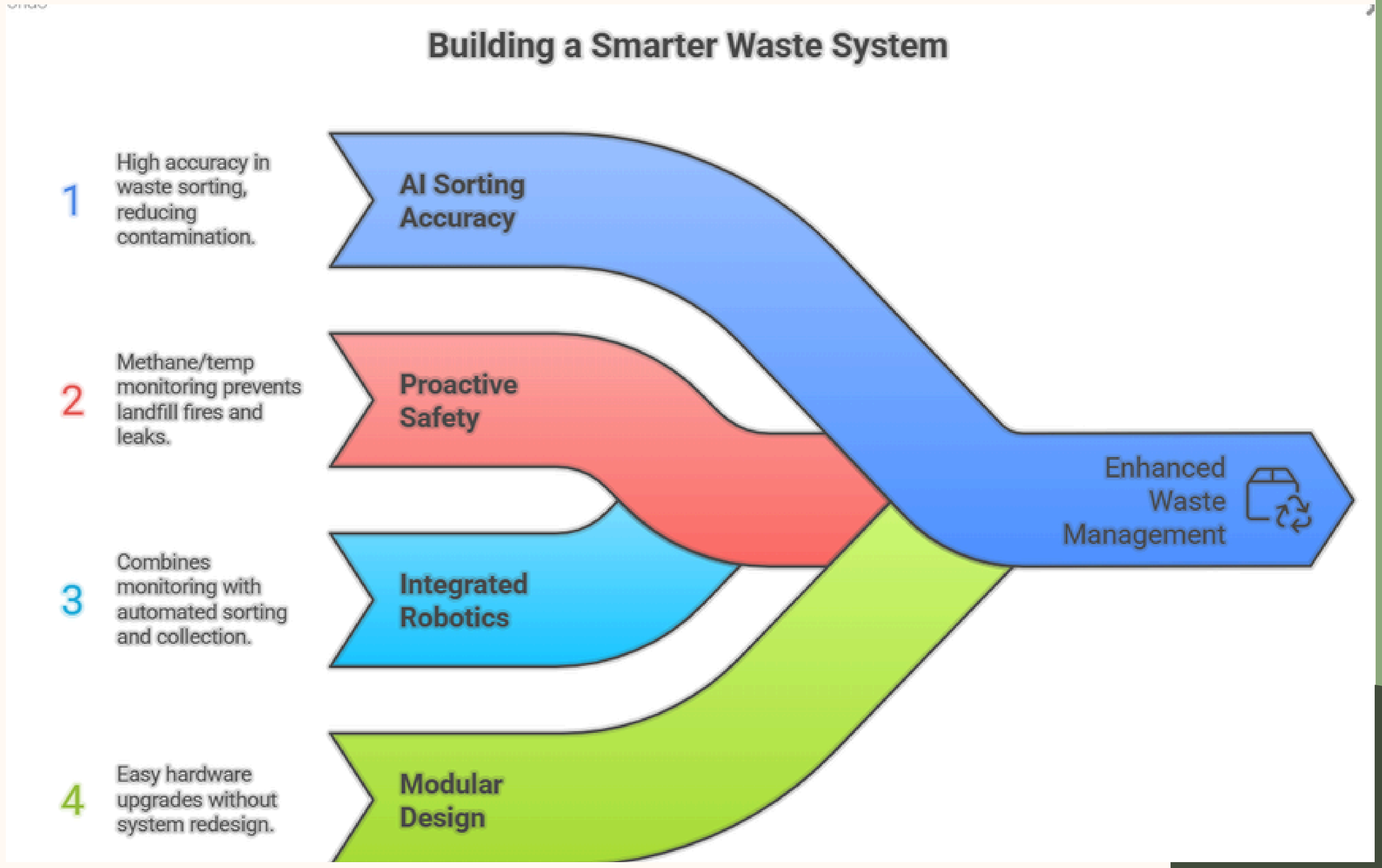
SOLUTION & UNIQUENESS

*98% AI Sorting Accuracy:*Dramatically reduces recycling contamination compared to manual methods.

Proactive Safety: First-of-its-kind methane/temp monitoring to prevent landfill fires and hazardous gas leaks.

Integrated Robotics: Combines monitoring with an automated 4-DOF sorting arm and GPS-guided collection vehicle.

Modular Design: Allows hardware upgrades (new sensors or AI models) without system redesign.



MARKET FEASIBILITY & APPLICATIONS

Smart Cities: *High-traffic urban zones like transit hubs and parks.*

Industrial Facilities: *Factories and campuses requiring safe handling of hazardous or metallic materials.*

Healthcare: *Adapting sensors for biomedical waste monitoring in hospitals.*

Rural/Remote Areas: *Off-grid, solar-powered bins for communities with limited electrification.*

THANK YOU!